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Education

- 2024 📖 **PhD degree, Peking University** in Space Physics
- 2022 – 2024 📖 **Visiting PhD student, IRAP, Toulouse, France** in Space Physics
Research field: *Origin and evolution of Alfvénic switchbacks.*
- 2019 – 2024 📖 **PhD student, Peking University** in Space Physics
Research field: *Dynamics of the solar atmosphere, Evolution of solar wind, Magnetic connectivity, Waves and turbulence in space plasma.*
- 2019 📖 **Bachelor's degree, Peking University** in Space Physics
- 2015 – 2019 📖 **Undergraduate, Peking University** in Space Physics

Fellowships, Awards and Scholarships

Fellowships

- 2025 📖 **Alexander von Humboldt Research Fellowship**, Alexander von Humboldt-Stiftung

Awards

- 2024 📖 **Best Student Paper Awards**, National Planetary Science Conference.
- 2023 📖 **Merit Student**, Peking University.
- 2022 📖 **Best Student Paper Awards**, Chinese Geoscience Union (CGU).
📖 **Best Student Poster Awards**, Asia Oceania Geosciences Society (AOGS).
- 2021 📖 **Graduate Award for Scientific Research**, Peking University.

Scholarships

- 2023 📖 **Peking University President's Scholarship**, Peking University.
- 2022 📖 **Chinese Government Scholarship**, China Scholarship Council.

Conference Organization

- 2026 📖 **Co-convenor**, Session ST29: Linking Plasma Dynamics in the Lower Solar Atmosphere to Magnetic Activity, AOGS Annual Meeting

Conference Presentations

Invited talks

- 2026  The Contribution of Turbulence to Solar Wind Heating in the Inner Heliosphere, 2nd Plasma Astrophysics School and Workshop (PASW26).

Oral presentation

- 2024  Evolution of Interplanetary Velocity Spikes and Their Contribution to Solar Wind Acceleration and Heating, National Planetary Science Conference.
- 2023  Possible role of fluctuation excitation in the formation of alfvénic fluctuations originating from interchange magnetic reconnection, European Geosciences Union (EGU).

Poster presentation

-  Connecting solar wind velocity spikes measured by solar orbiter and coronal bright points imaged by SDO, Solar Wind 16 meeting.
-  Jet-flow fluctuations and plasma blobs as a mediator between interchange magnetic reconnection in solar corona and Alfvénic velocity spikes in interplanetary space, Asia Oceania Geosciences Society (AOGS).
- 2022  From magnetic reconnection at chromospheric network boundaries to switchbacks in the inner heliosphere, Asia Oceania Geosciences Society (AOGS).

Research Publications

Journal Articles (†, co-first author)

- 1 **C. Hou**, J. He, A. Rouillard, et al., “Non-adiabatic entropy increase in the solar wind governed by velocity spikes”, *Nature Astronomy*, accepted in principle (2026).
- 2 **C. Hou**, H. Yan, and S. Zhao, “Compressible turbulence as a source of particle beams and ion Bernstein waves in collisionless plasmas”, *The Astrophysical Journal*, accepted (2026).
- 3 **C. Hou**, H. Yan, S. Zhao, and P. Pavaskar, “Energy cascade and damping in fast-mode compressible turbulence”, *The Astrophysical Journal Letters* **992**, L28 (2025).
- 4 **C. Hou**†, B. Gannouni†, A. P. Rouillard, J. He, and V. Réville, “Fine-scale activity driven by magnetic reconnection within coronal microjets”, *Astronomy & Astrophysics* **697**, A67 (2025).
- 5 **C. Hou**, J. He, D. Duan, Z. Wu, Y. Chen, D. Verscharen, A. P. Rouillard, H. Li, L. Yang, and S. D. Bale, “The origin of interplanetary switchbacks in reconnection at chromospheric network boundaries”, *Nature Astronomy* **8**, 1246–1256 (2024).
- 6 **C. Hou**, A. P. Rouillard, J. He, B. Gannouni, V. Réville, P. Louarn, A. Fedorov, L. Přech, C. J. Owen, D. Verscharen, R. D’Amicis, L. Sorriso-Valvo, N. Fargette, J. Coburn, V. Génot, J. M. Raines, R. Bruno, S. Livì, B. Lavraud, N. André, G. Fruit, R. Kieokaew, I. Plotnikov, E. Penou, A. Barthe, D. Kataria, M. Berthomier, F. Allegrini, V. Fortunato, G. Mele, and T. Horbury, “Connecting solar wind velocity spikes measured by solar orbiter and coronal brightenings observed by sdo”, *The Astrophysical Journal Letters* **968**, L28 (2024).

- 7 **C. Hou**[†], X. Zhu[†], R. Zhuo[†], J. He, D. Verscharen, and D. Duan, “Nature, generation, and dissipation of alfvénic kinks/switchbacks observed by parker solar probe and wind”, *The Astrophysical Journal* **950**, 157 (2023).
- 8 **C. Hou**, J. He, D. Duan, X. Zhu, W. Li, D. Verscharen, T. Liu, and T. Wang, “Efficient energy conversion through vortex arrays in the turbulent magnetosheath”, *The Astrophysical Journal* **946**, 13 (2023).
- 9 **C. Hou**, J. He, X. Zhu, and Y. Wang, “Contribution of magnetic reconnection events to energy dissipation in space plasma turbulence”, *The Astrophysical Journal* **908**, 237 (2021).
- 10 **C. Hou**, J. He, L. Zhang, Y. Wang, and D. Duan, “Dynamics of the charged particles released from a sun-grazing comet in the solar corona”, *Earth and Planetary Physics* **5**, 232–238 (2021).
- 11 Y. Sun[†], J. Zhao[†], **C. Hou**[†], and W. Jiao, “Highlight advances in planetary physics in the solar system: in situ detection over the past 20 years”, *Space: Science & Technology* **3**, 0007 (2023).
- 12 L. Yang, **C. Hou**, X. Feng, J. He, M. Xiong, M. Zhang, Y. Zhou, F. Shen, X. Zhao, H. Li, et al., “Global morphology distortion of the 2021 october 9 coronal mass ejection from an ellipsoid to a concave shape”, *The Astrophysical Journal* **942**, 65 (2023).
- 13 Z. Wu, J. He, **C. Hou**, D. Duan, J. Huang, A. P. Rouillard, D. Verscharen, Y. Chen, R. Zhuo, and T. Chen, “Multiscale magnetic reconnection in the genesis of young slow solar wind”, *The Astrophysical Journal Supplement Series* **282**, 4 (2026).